WEEK 5 – Power BI

Healthcare Data Cleaning & Data Modeling

- **APARNA F**
  B.Sc Data Science & AI

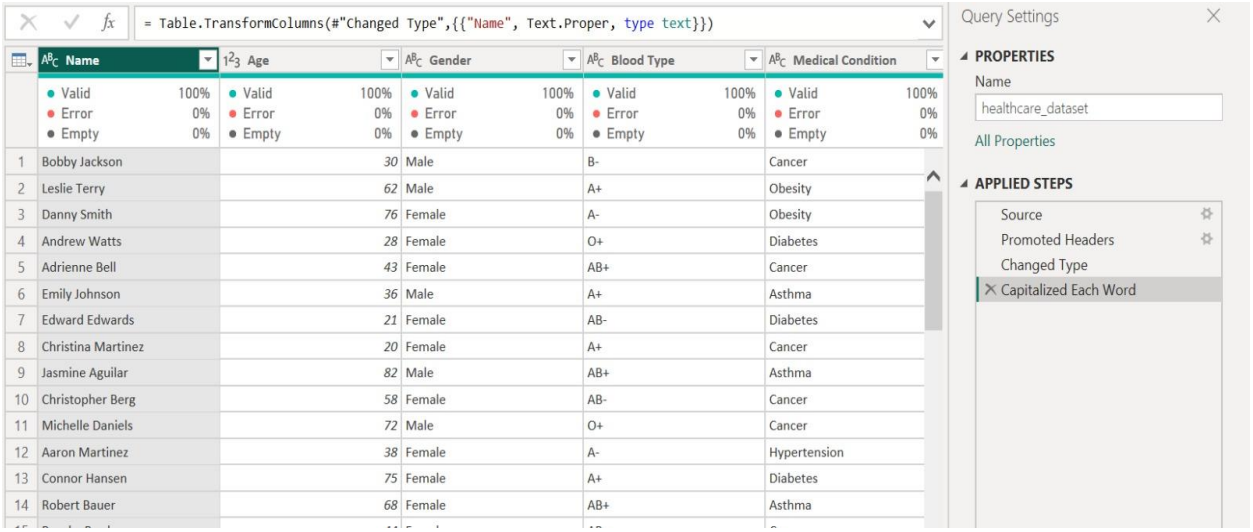
Introduction

Week 5 focuses on **cleaning, transforming, and organizing the healthcare dataset using Power BI**. The raw dataset is first imported into Power Query, where the data is inspected for **missing values, errors, duplicate records, incorrect data types, and inconsistencies**. Text fields are standardized, date and numerical columns are assigned appropriate data types, and required derived fields such as **Age Group, Patient ID, and Admission ID** are created to improve data organization.

After completing the cleaning process, the data is structured into **dimension and fact tables**, such as **Dim_Patient, Dim_Admission, and Fact_Billing**. Relationships are then established between these tables using appropriate keys such as **Patient ID and Admission ID**, creating a structured **star-schema data model**. This prepares the healthcare dataset for accurate and efficient analysis in the later stages of the Power BI project.

Capitalizing Each Word

The **Capitalized Each Word** transformation was applied to the **Name** column in Power Query. This standardizes the formatting of patient names by converting the first letter of each word to uppercase, making the data consistent and easier to read. The transformation was applied using **Transform → Format → Capitalize Each Word**, and the change appears under **Applied Steps** in Power Query.

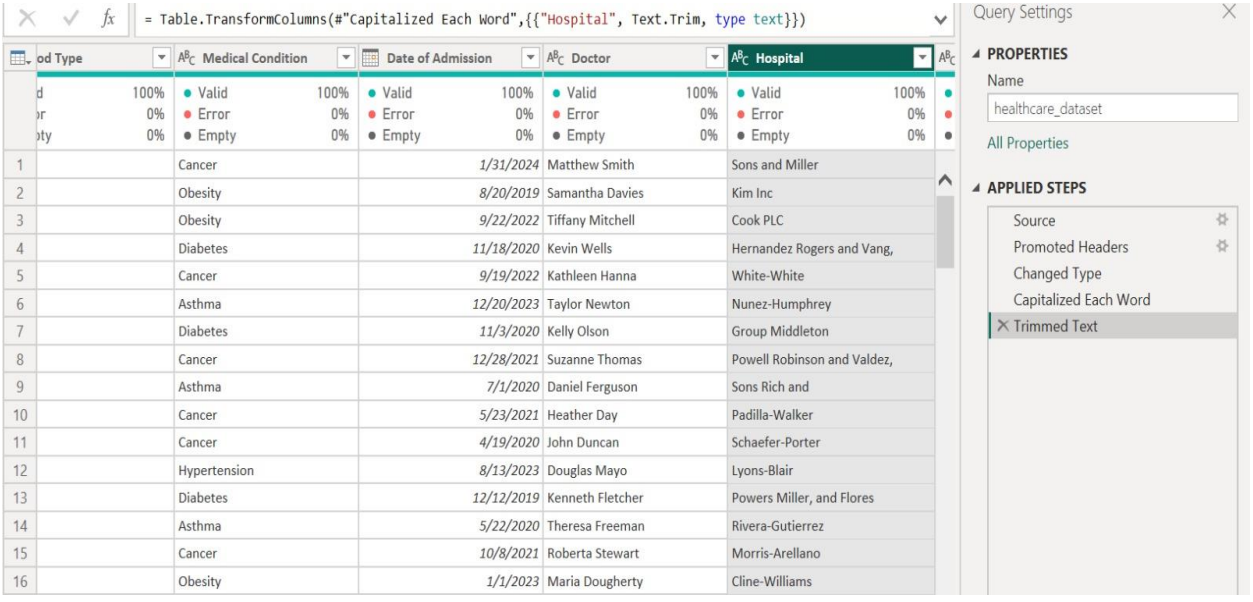


	Name	Age	Gender	Blood Type	Medical Condition
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	Bobby Jackson	30	Male	B-	Cancer
2	Leslie Terry	62	Male	A+	Obesity
3	Danny Smith	76	Female	A-	Obesity
4	Andrew Watts	28	Female	O+	Diabetes
5	Adrienne Bell	43	Female	AB+	Cancer
6	Emily Johnson	36	Male	A+	Asthma
7	Edward Edwards	21	Female	AB-	Diabetes
8	Christina Martinez	20	Female	A+	Cancer
9	Jasmine Aguilar	82	Male	AB+	Asthma
10	Christopher Berg	58	Female	AB-	Cancer
11	Michelle Daniels	72	Male	O+	Cancer
12	Aaron Martinez	38	Female	A-	Hypertension
13	Connor Hansen	75	Female	A+	Diabetes
14	Robert Bauer	68	Female	AB+	Asthma

Trimmed Text

The **Trimmed Text** transformation was applied to the **Hospital** column in Power Query. This removes unnecessary spaces from the beginning and end of hospital names, ensuring consistent and clean text data.

The transformation was applied using **Transform → Format → Trim**, and the **Trimmed Text** step can be seen in the **Applied Steps** section. This helps maintain consistency when grouping, filtering, or analyzing hospital information.



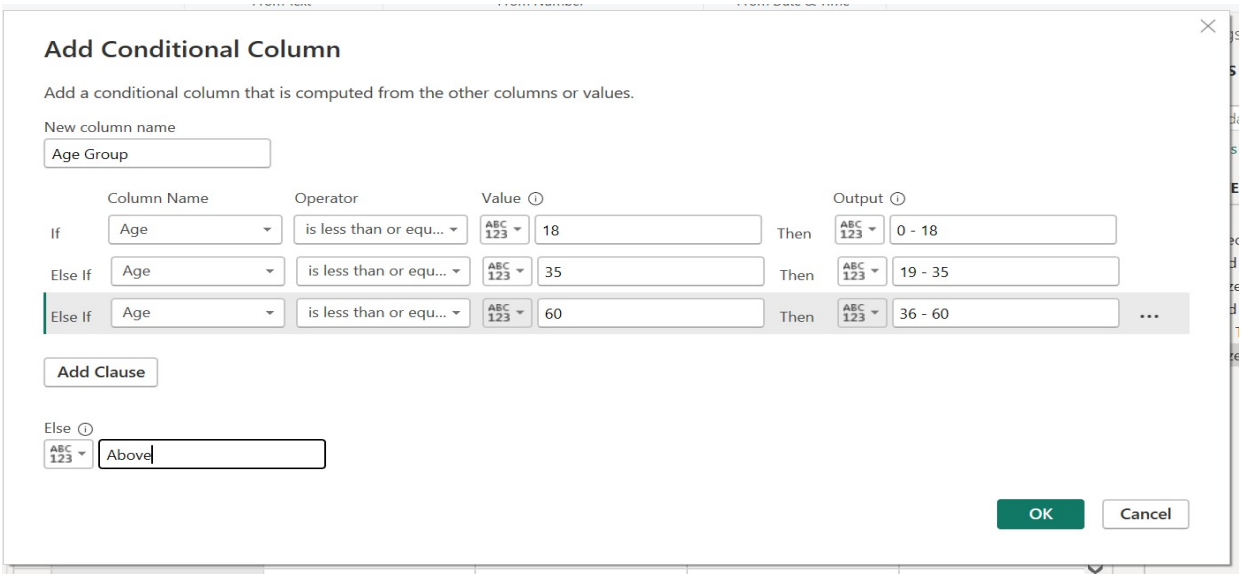
	od Type	Medical Condition	Date of Admission	Doctor	Hospital
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1		Cancer	1/31/2024	Matthew Smith	Sons and Miller
2		Obesity	8/20/2019	Samantha Davies	Kim Inc
3		Obesity	9/22/2022	Tiffany Mitchell	Cook PLC
4		Diabetes	11/18/2020	Kevin Wells	Hernandez Rogers and Vang,
5		Cancer	9/19/2022	Kathleen Hanna	White-White
6		Asthma	12/20/2023	Taylor Newton	Nunez-Humphrey
7		Diabetes	11/3/2020	Kelly Olson	Group Middleton
8		Cancer	12/28/2021	Suzanne Thomas	Powell Robinson and Valdez,
9		Asthma	7/1/2020	Daniel Ferguson	Sons Rich and
10		Cancer	5/23/2021	Heather Day	Padilla-Walker
11		Cancer	4/19/2020	John Duncan	Schaefer-Porter
12		Hypertension	8/13/2023	Douglas Mayo	Lyons-Blair
13		Diabetes	12/12/2019	Kenneth Fletcher	Powers Miller, and Flores
14		Asthma	5/22/2020	Theresa Freeman	Rivera-Gutierrez
15		Cancer	10/8/2021	Roberta Stewart	Morris-Arellano
16		Obesity	1/1/2023	Maria Dougherty	Cline-Williams

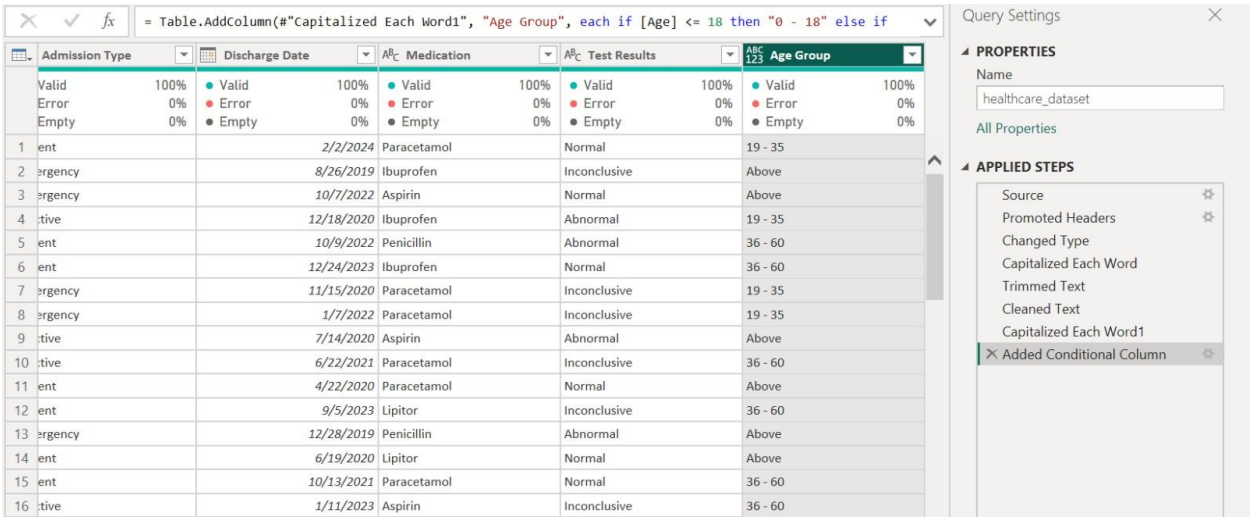
Age Group – Conditional Column

A **Conditional Column** named **Age Group** was created in Power Query to categorize patients based on their age. The age values were divided into meaningful demographic groups: **0–18, 19–35, 36–60, and Above 60**.

The conditions shown in the image classify each patient according to the `Age` column. The **Else** condition is set to **"Above 60"**, so patients whose age is greater than 60 are placed in this group.

This transformation makes the patient data easier to **categorize, filter, and analyze** based on age groups.



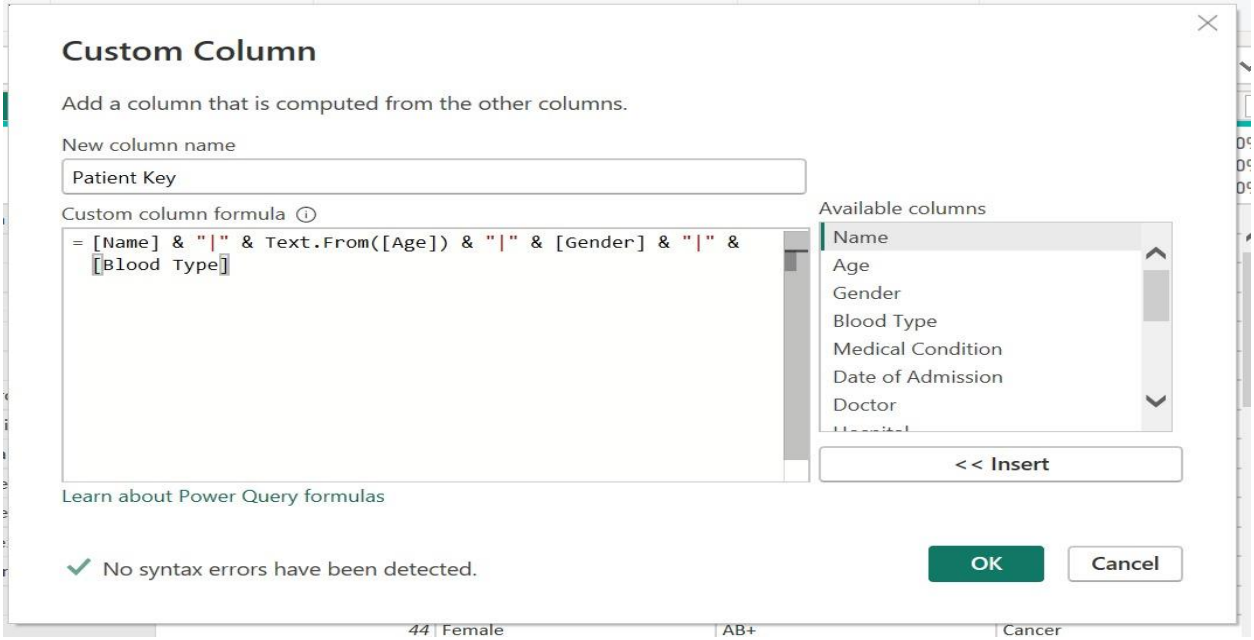


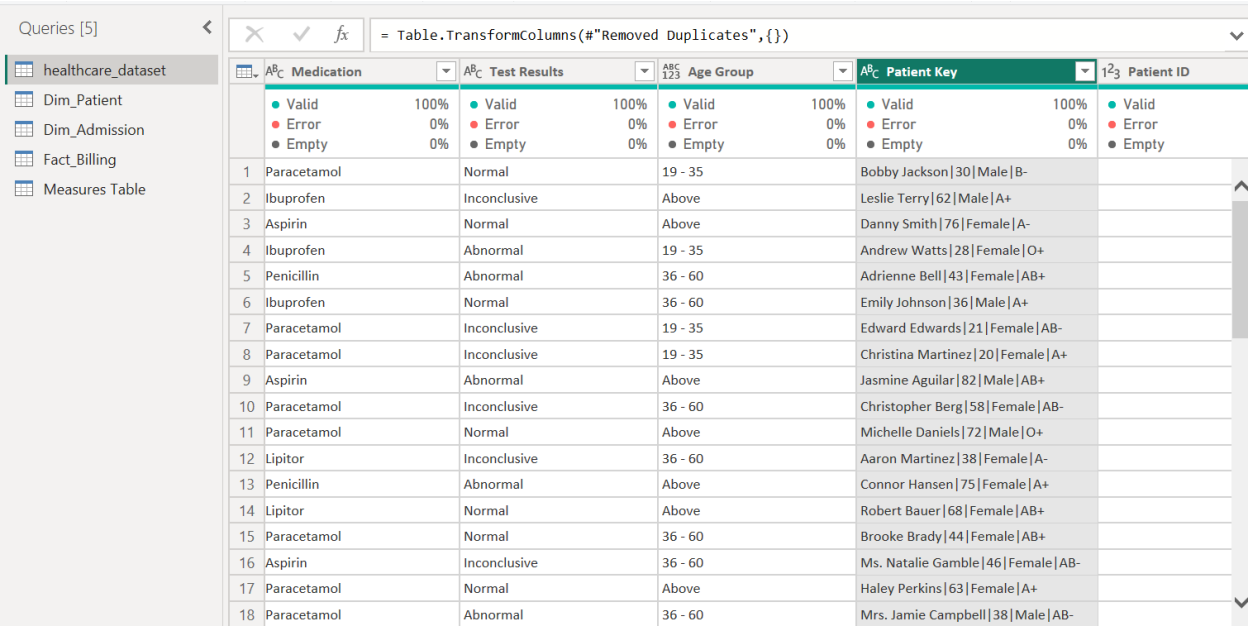
Patient Key – Custom Column

A **Patient Key** custom column was created in Power Query to uniquely identify and match patient records. It combines the patient's **Name, Age, Gender, and Blood Type** into a single text value, helping to organize the patient data.

Formula:

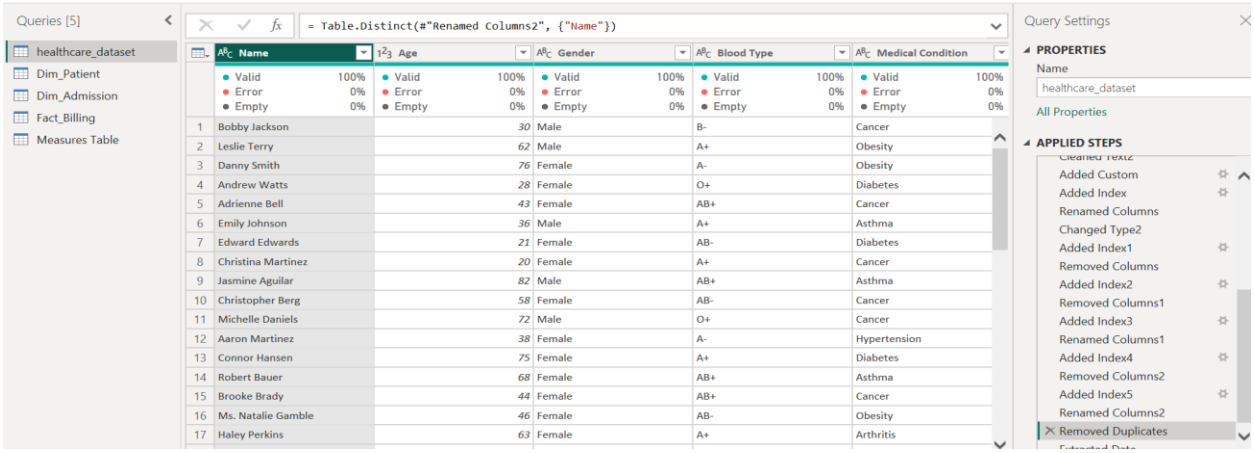
[Name] & "|" & Text.From([Age]) & "|" & [Gender] & "|" & [Blood Type]





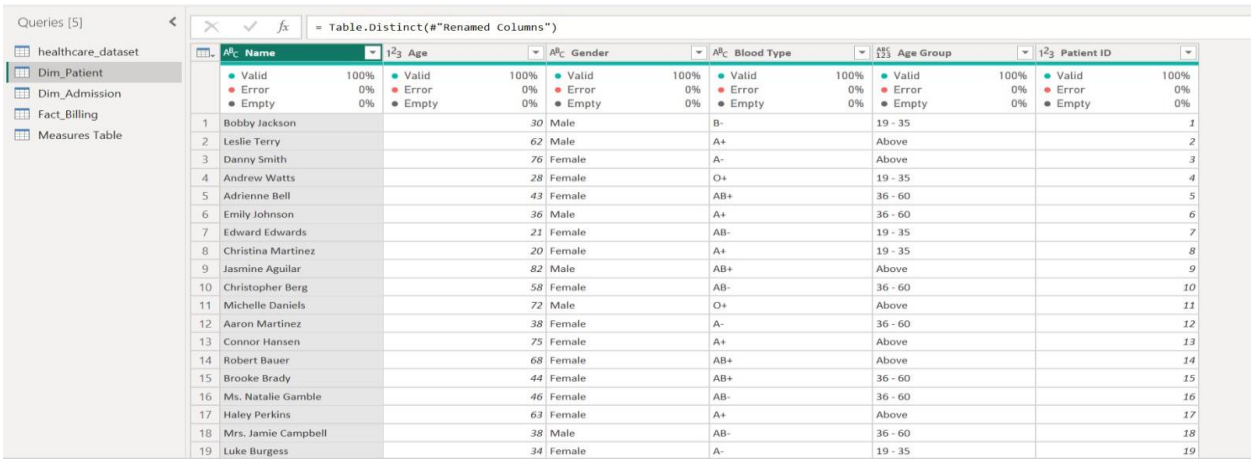
Remove Duplicates

The **Remove Duplicates** transformation was applied to the **Name** column in Power Query to ensure that each patient appears only once in the patient dimension table. Duplicate patient records were identified and removed, helping to maintain **unique patient information and data consistency**. This step is important when creating the `Dim_Patient` table because it ensures that each patient can be assigned a unique **Patient ID** and used correctly in the data model.



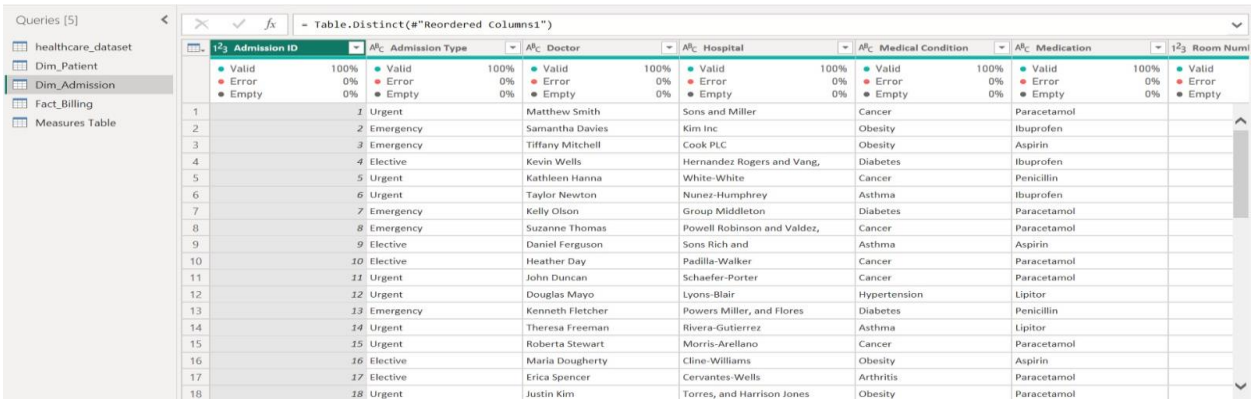
Creating Dim_Patient Table

A **Dim_Patient** table was created in Power Query to store unique and organized patient information. The table contains **Name, Age, Gender, Blood Type, Age Group, and Patient ID**. Duplicate patient records were removed, and a unique **Patient ID** was assigned to each patient. This table provides a structured view of patient details and can be connected to the **Fact_Billing** table through the Patient ID for further analysis.



Creating Dim_Admission Table

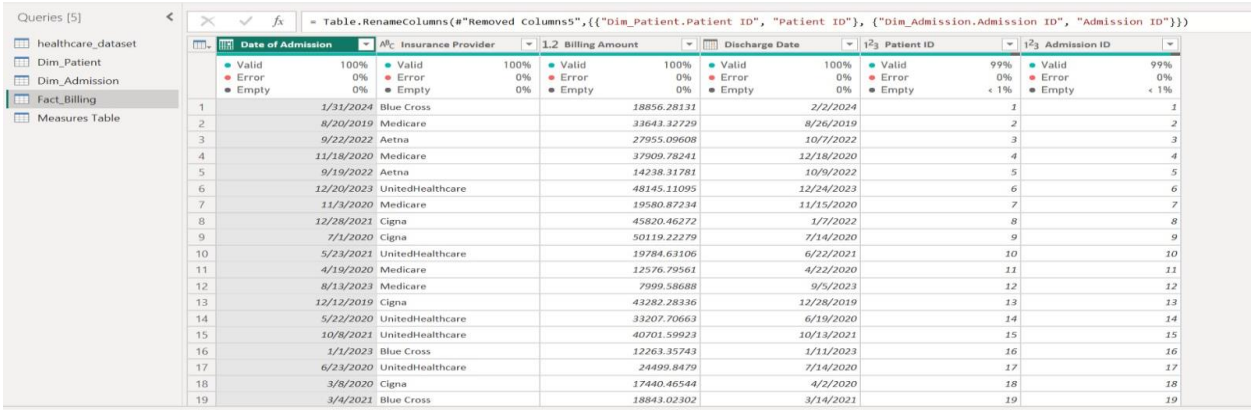
A **Dim_Admission** table was created in Power Query to organize and store admission-related information separately from the main healthcare dataset. The table includes **Admission ID, Admission Type, Doctor, Hospital, Medical Condition, Medication, and Room Number**. Duplicate records were removed to maintain unique admission details, and the **Admission ID** is used as the key for identifying individual admission records. This table helps create a structured data model and can be connected to the **Fact_Billing** table using Admission ID.



	Admission ID	Admission Type	Doctor	Hospital	Medical Condition	Medication	Room Numl
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid Error Empty
1	1	Urgent	Matthew Smith	Sons and Miller	Cancer	Paracetamol	
2	2	Emergency	Samantha Davies	Kim Inc	Obesity	Ibuprofen	
3	3	Emergency	Tiffany Mitchell	Cook PLC	Obesity	Aspirin	
4	4	Elective	Kevin Wells	Hernandez Rogers and Vang,	Diabetes	Ibuprofen	
5	5	Urgent	Kathleen Hanna	White-White	Cancer	Penicillin	
6	6	Urgent	Taylor Newton	Nunez-Humphrey	Asthma	Ibuprofen	
7	7	Emergency	Kelly Olson	Group Middleton	Diabetes	Paracetamol	
8	8	Emergency	Suzanne Thomas	Powell Robinson and Valdez,	Cancer	Paracetamol	
9	9	Elective	Daniel Ferguson	Sons Rich and	Asthma	Aspirin	
10	10	Elective	Heather Day	Padilla-Walker	Cancer	Paracetamol	
11	11	Urgent	John Duncan	Schaefer-Porter	Cancer	Paracetamol	
12	12	Urgent	Douglas Mayo	Lyons-Blair	Hypertension	Lipitor	
13	13	Emergency	Kenneth Fletcher	Powers Miller, and Flores	Diabetes	Penicillin	
14	14	Urgent	Theresa Freeman	Rivera-Gutierrez	Asthma	Lipitor	
15	15	Urgent	Roberta Stewart	Morris-Arellano	Cancer	Paracetamol	
16	16	Elective	Maria Dougherty	Cline-Williams	Obesity	Aspirin	
17	17	Elective	Erica Spencer	Cervantes-Wells	Arthritis	Paracetamol	
18	18	Urgent	Justin Kim	Torres, and Harrison Jones	Obesity	Paracetamol	

Creating Fact_Billing Table

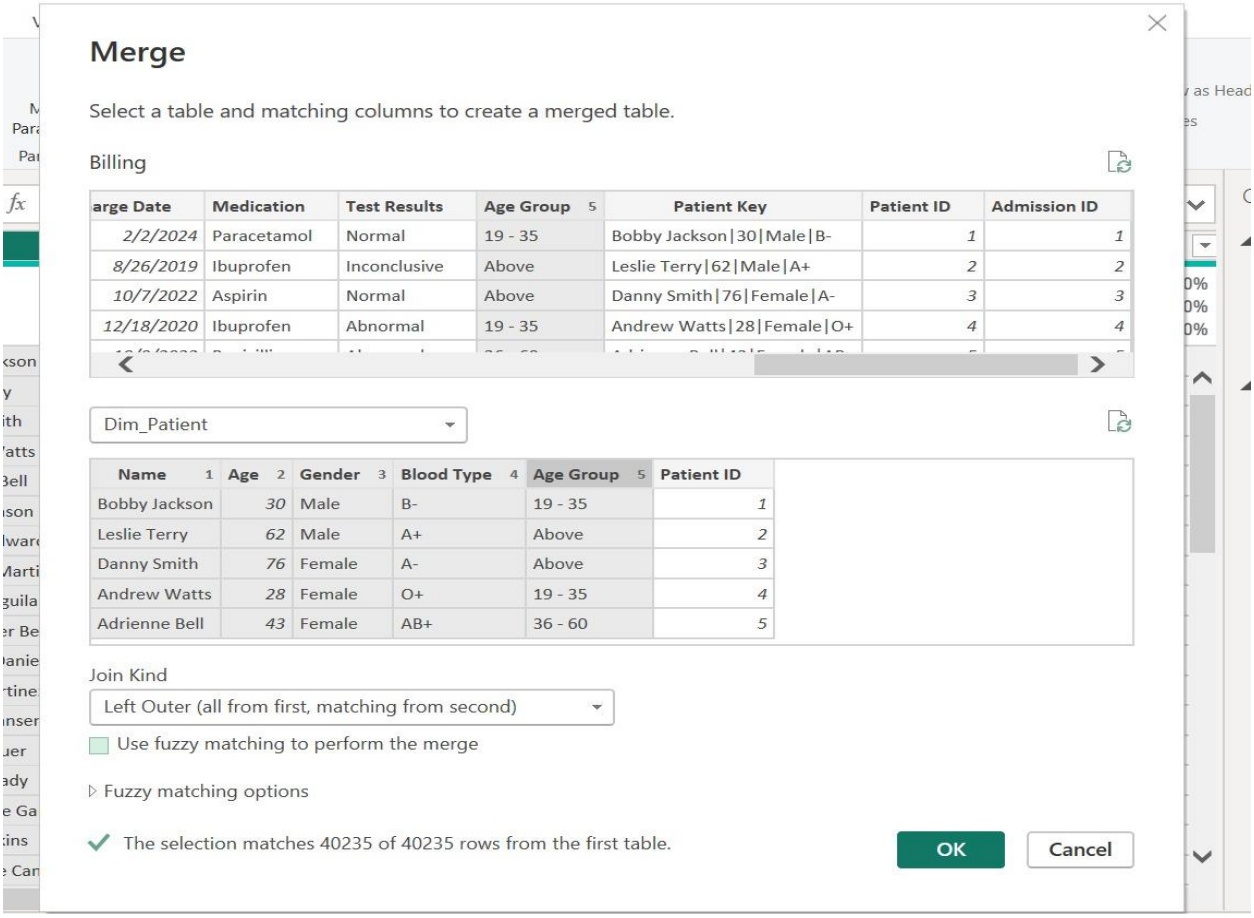
The **Fact_Billing** table was created to store the main billing and admission-related transaction data. It contains important fields such as **Date of Admission, Insurance Provider, Billing Amount, Discharge Date, Patient ID, and Admission ID**. The **Patient ID** and **Admission ID** act as keys that connect the billing records with the **Dim_Patient** and **Dim_Admission** tables. This structure helps maintain a proper relationship between patient, admission, and billing information and supports accurate analysis in the Power BI dashboard.



	Date of Admission	Insurance Provider	Billing Amount	Discharge Date	Patient ID	Admission ID
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 99% Error 0% Empty < 1%	Valid 99% Error 0% Empty < 1%
1	1/31/2024	Blue Cross	18856.28131	2/2/2024	1	1
2	8/20/2019	Medicare	33643.32729	8/26/2019	2	2
3	9/22/2022	Aetna	27955.09608	10/7/2022	3	3
4	11/18/2020	Medicare	37909.78241	12/18/2020	4	4
5	9/19/2022	Aetna	14238.31781	10/9/2022	5	5
6	12/20/2023	UnitedHealthcare	48145.11095	12/24/2023	6	6
7	11/3/2020	Medicare	19580.87234	11/15/2020	7	7
8	12/28/2021	Cigna	45820.46272	1/7/2022	8	8
9	7/1/2020	Cigna	50119.22279	7/14/2020	9	9
10	5/23/2021	UnitedHealthcare	19784.63106	6/22/2021	10	10
11	4/19/2020	Medicare	12576.79561	4/22/2020	11	11
12	8/13/2023	Medicare	7999.58688	9/5/2023	12	12
13	12/12/2019	Cigna	43282.28336	12/28/2019	13	13
14	5/22/2020	UnitedHealthcare	33207.70663	6/19/2020	14	14
15	10/8/2021	UnitedHealthcare	40701.59923	10/13/2021	15	15
16	1/1/2023	Blue Cross	12263.35743	1/11/2023	16	16
17	6/23/2020	UnitedHealthcare	24499.8479	7/14/2020	17	17
18	3/8/2020	Cigna	17440.46544	4/2/2020	18	18
19	3/4/2021	Blue Cross	18843.02302	3/14/2021	19	19

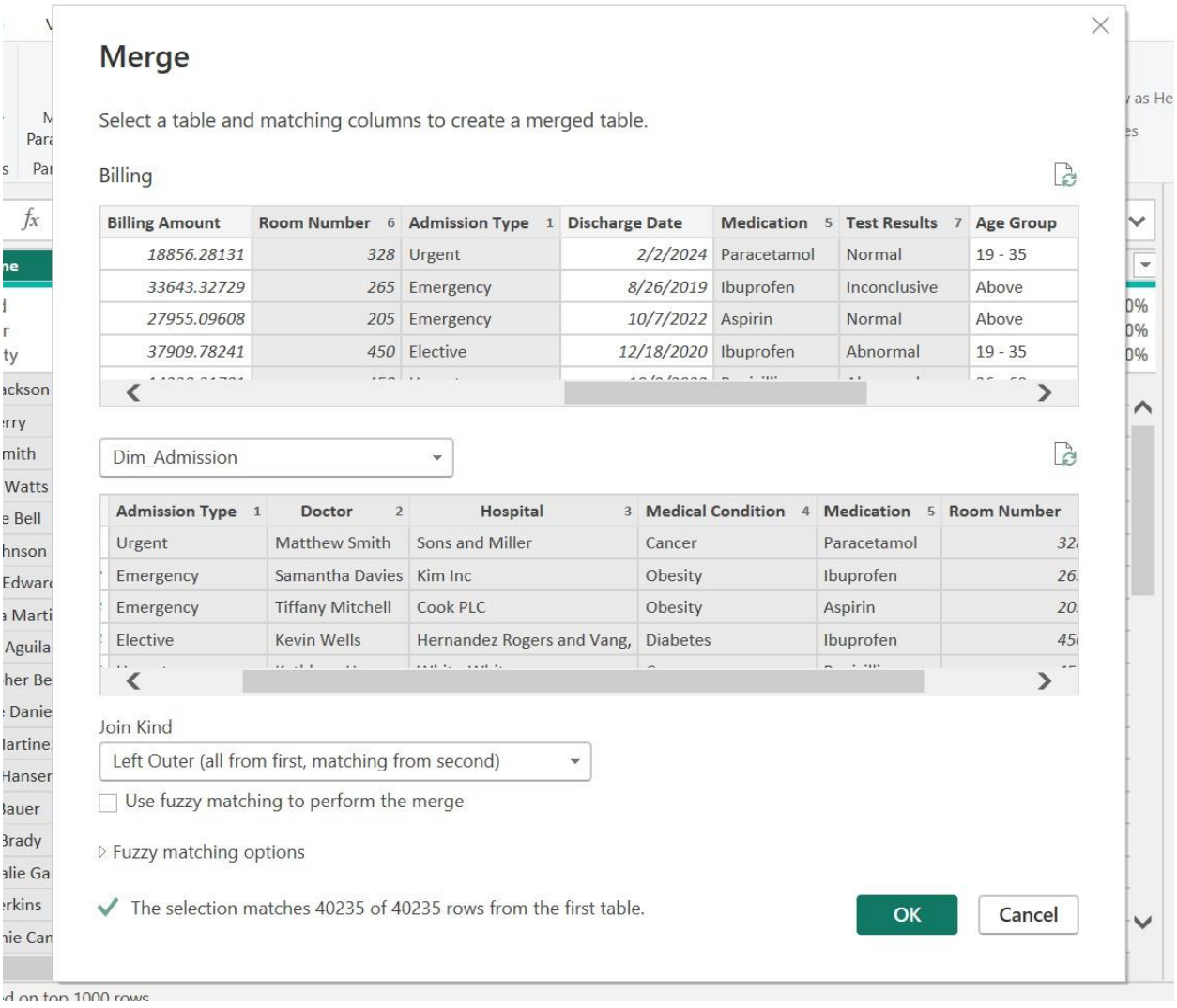
Merging healthcare_dataset with Dim_Patient

The **healthcare_dataset** was merged with the **Dim_Patient** table using the **Patient ID** column as the matching key. A **Left Outer Join** was selected so that all records from the healthcare dataset were retained while matching patient details were obtained from `Dim_Patient`. The merge matched **40,235 out of 40,235 rows**, confirming that the patient IDs were correctly matched. This helped connect the main healthcare records with the patient dimension table for further analysis.



Merging Healthcare Dataset with Dim_Admission

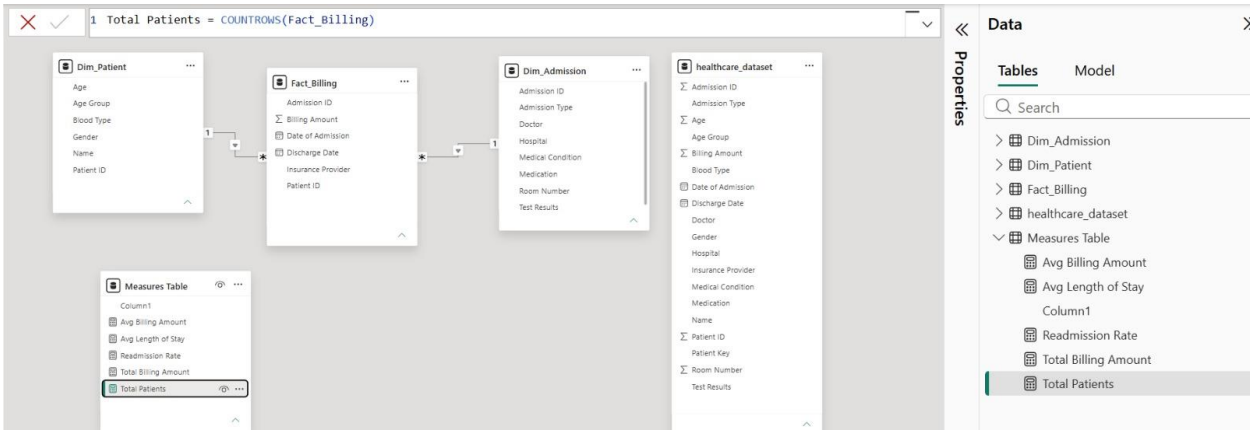
The **Healthcare Dataset** was merged with the **Dim_Admission** table in Power Query using matching admission-related columns. A **Left Outer Join** was selected so that all records from the Healthcare Dataset were retained while matching admission details were added from `Dim_Admission`. The merge showed that **40,235 out of 40,235 rows matched**, confirming a successful connection between the two tables.



Total Patients

The **Total Patients** measure calculates the total number of records available in the **Fact_Billing** table. The `COUNTROWS()` function counts each row in the table, with each row representing a billing/admission record. This measure can be used in a **KPI card** to show the total number of patient records in the dataset.

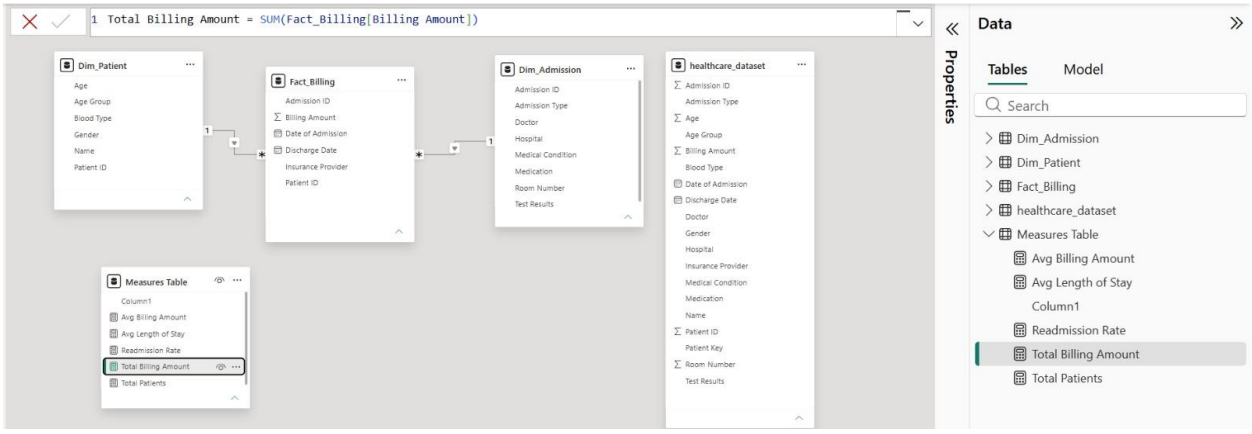
DAX Formula: `Total Patients = COUNTROWS(Fact_Billing)`



Total Billing Amount

The **Total Billing Amount** measure calculates the total amount billed across all records in the **Fact_Billing** table. The `SUM()` function adds all the values in the **Billing Amount** column. This measure is useful for understanding the overall billing revenue and can be displayed in a **KPI card** or used in charts for further analysis.

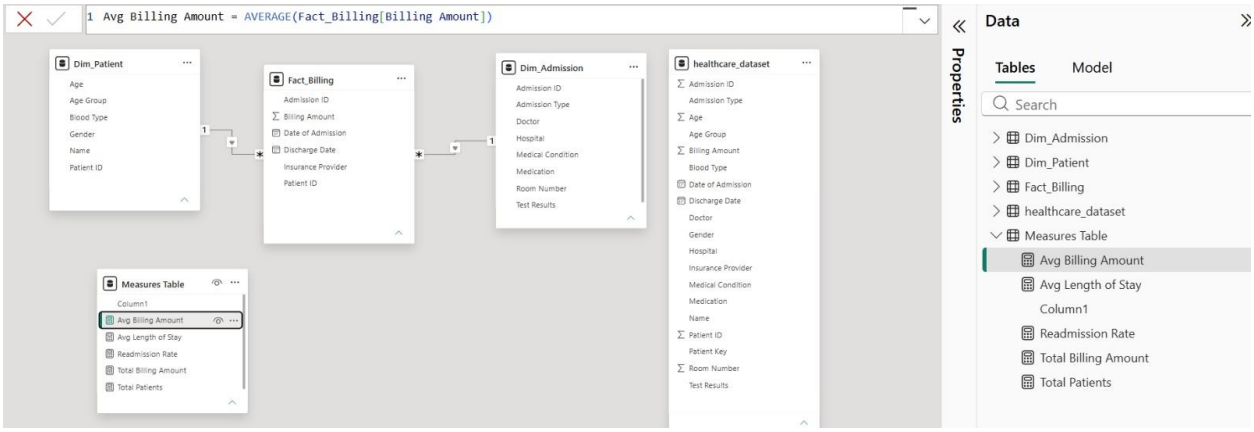
DAX Formula:`Total Billing Amount = SUM(Fact_Billing[Billing Amount])`



Average Billing Amount

The **Average Billing Amount** measure calculates the average amount billed for all patient billing records. It is useful for understanding the typical billing amount and can be displayed in a **KPI card** or used in other dashboard visuals.

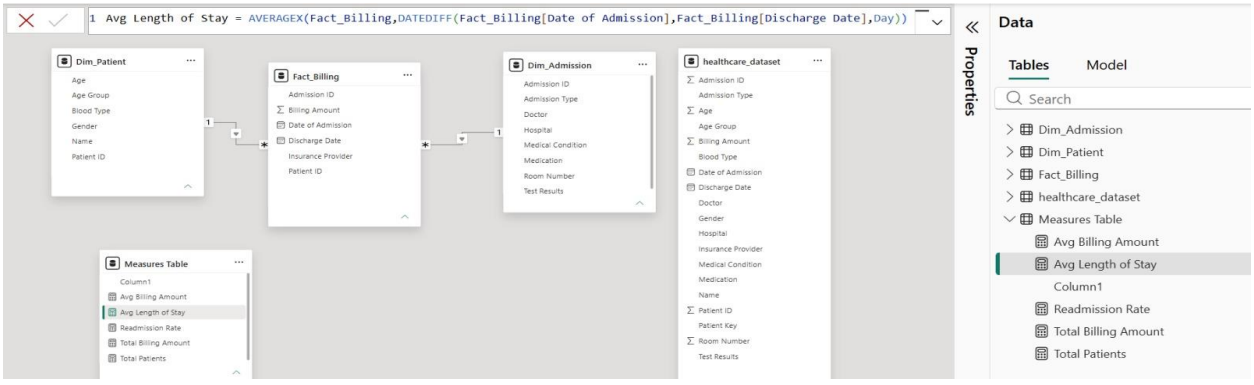
DAX Formula: `Avg Billing Amount = AVERAGE(Fact_Billing[Billing Amount])`



Average Length of Stay

The **Average Length of Stay** measure calculates the average number of days patients stayed in the hospital. It finds the number of days between the **Date of Admission** and **Discharge Date** for each patient and then calculates the average across all records. This measure is useful for analyzing hospital stay duration and can be displayed in a **KPI card** or dashboard visual.

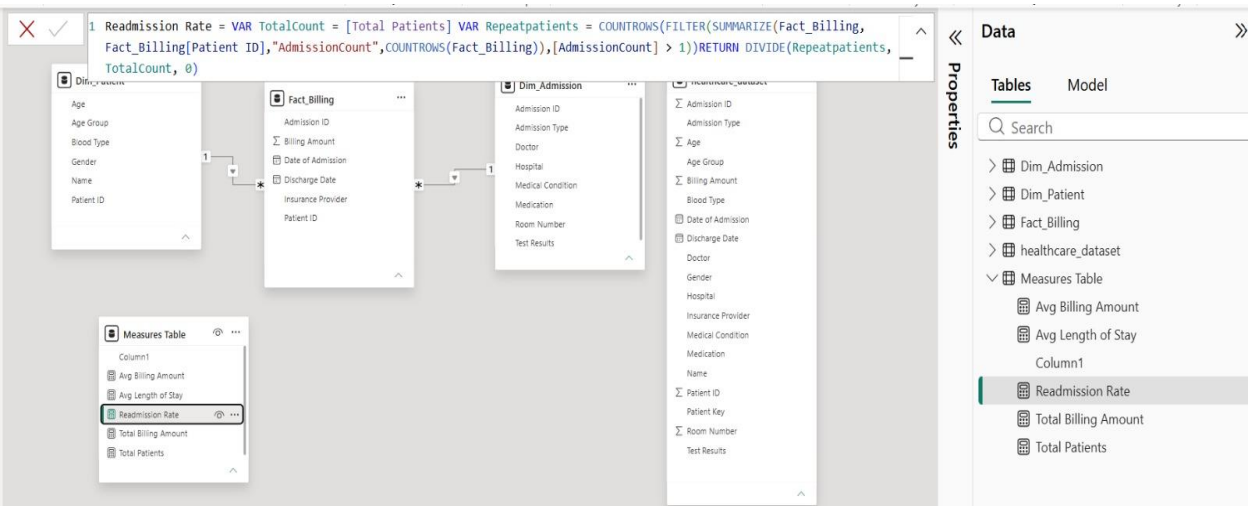
DAX Formula:
```
Avg Length of Stay =
AVERAGEX(Fact_Billing,DATEDIFF(Fact_Billing[Date of
Admission],Fact_Billing[Discharge Date],Day))
```



Readmission Rate

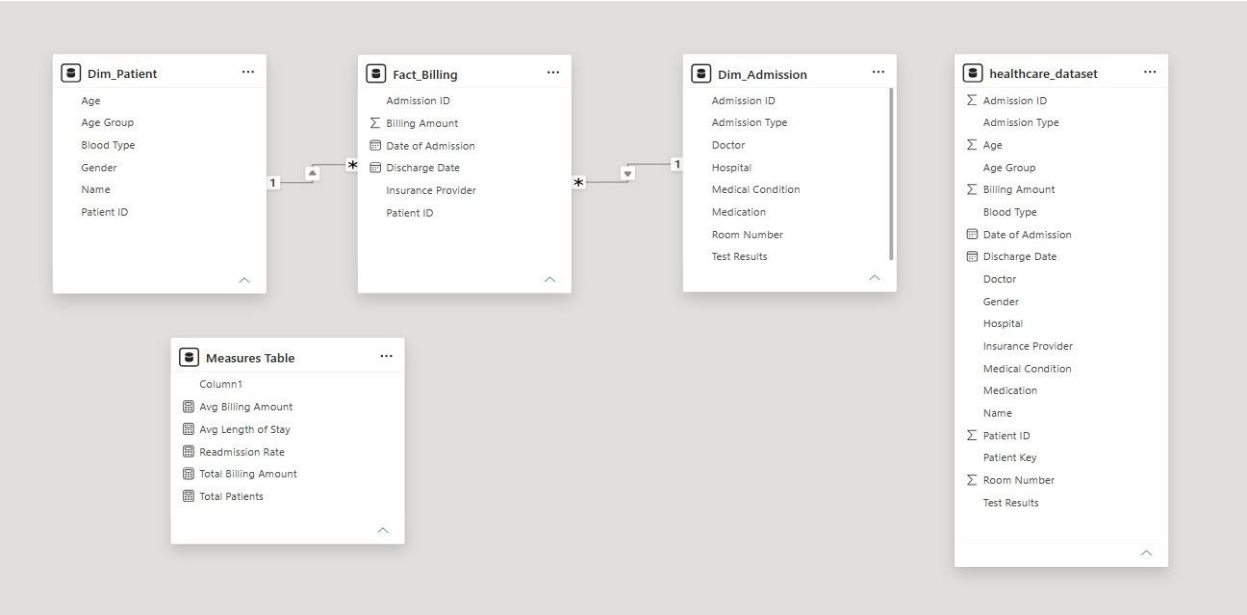
The **Readmission Rate** measure calculates the percentage of patients who have been admitted more than once. The formula groups the billing records by **Patient ID**, counts the number of admissions for each patient, identifies patients with more than one admission, and then divides the number of repeat patients by the total number of patients. This measure helps to understand the proportion of patients with repeated admissions.

DAX Formula: `Readmission Rate = VAR TotalCount = [Total Patients] VAR Repeatpatients = COUNTROWS(FILTER(SUMMARIZE(Fact_Billing,Fact_Billing[Patient ID],"AdmissionCount",COUNTROWS(Fact_Billing)),[AdmissionCount] > 1))RETURN DIVIDE(Repeatpatients,TotalCount, 0)`



Relationship Creation:

Relationships were created in Power BI to establish connections between the fact and dimension tables. The Fact_Billing table was connected to Dim_Patient using the **Patient ID** column and to Dim_Admission using the **Admission ID** column. Both relationships were configured as **Many-to-One (*:1)** relationships and were kept active. The dimension tables contain unique ID values, while the fact table contains multiple transaction records associated with those IDs. These relationships enable Power BI to correctly filter and analyze billing data based on patient and admission attributes, providing a structured foundation for the healthcare dashboard.



Conclusion

Overall, Week 5 focused on **cleaning and preparing the healthcare dataset and establishing relationships in Power BI**. The data was cleaned by correcting data types, formatting text, removing duplicates, creating useful columns such as **Age Group and Patient ID**, and organizing the data into **Dim_Patient, Dim_Admission, and Fact_Billing** tables. Appropriate relationships were then created between the tables using **Patient ID and Admission ID**. DAX measures such as **Total Patients, Total Billing Amount, Average Billing Amount, Average Length of Stay, and Readmission Rate** were also created for analysis. These steps provided a structured and reliable data model for developing the healthcare dashboard and performing meaningful analysis.